

FRD_Logistics_Gateway

Please note: The provided Business Requirement Document (BRD) does not contain a specific section titled "Coding Requirements". Therefore, I have used the entire provided BRD text to infer the functional requirements, as it represents the core business need and solution description.

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Functional Requirement Document (FRD)

1. FR-GW-001: Ingest Raw Sensor Data

- Title: Ingest Raw Sensor Data
- Description: The gateway system must be capable of receiving and buffering raw data streams from various IoT sensors installed on delivery trucks. This data includes updates about location, fuel, and engine health.
- Preconditions:
 - IoT sensors are active and transmitting data.
 - Gateway system is operational and connected to the sensor network.
 - Network connectivity between sensors and the gateway is established.
- Main Flow / Functional Steps:
 - The gateway continuously monitors for incoming data from connected IoT sensors.
 - Upon receiving data, the gateway buffers the "raw" sensor text.
 - The gateway identifies the source sensor for each incoming data packet.

2. FR-GW-002: Normalize Sensor Data Structure

- Title: Normalize Sensor Data Structure
- Description: The gateway must transform incoming raw sensor data, which may originate from different manufacturers (e.g., Bosch vs. Garmin) with varying data structures, into a predefined company-standard schema.
- Preconditions:
 - Raw sensor data has been successfully ingested (FR-GW-001 completed).
 - A company-standard data schema for location, fuel, and engine health is defined and accessible.
 - Mapping rules for different sensor manufacturers' data structures to the standard schema are configured within the gateway.
- Main Flow / Functional Steps:
 - The gateway retrieves a buffered raw sensor data packet.
 - It identifies the manufacturer and original data structure based on the sensor's metadata or data format.

- The gateway applies the corresponding pre-configured transformation rules to convert the raw data fields into the company's standard schema.
- The normalized data is then prepared for further processing.

3. FR-GW-003: Clean Noisy Sensor Data

- Title: Clean Noisy Sensor Data
- Description: The gateway must identify and clean or filter out "noisy" data points, such as those resulting from temporary signal loss (e.g., brief GPS signal loss), to ensure data quality before further processing or routing.
- Preconditions:
 - Raw sensor data has been successfully ingested (FR-GW-001 completed).
 - Data cleaning rules or algorithms for identifying and handling "noisy" data are defined and configured within the gateway.
- Main Flow / Functional Steps:
 - The gateway analyzes incoming raw or partially processed sensor data for anomalies indicative of "noisy" data (e.g., improbable location jumps, sudden drops in fuel level, inconsistent engine health readings).
 - It applies defined cleaning logic (e.g., discarding data points, flagging data as unreliable, interpolating missing values, smoothing data) to address the noisy data.
 - Cleaned data is then passed on for normalization (if not already normalized) or routing.

4. FR-GW-004: Route Engine Alerts to Maintenance Agent

- Title: Route Engine Alerts to Maintenance Agent
- Description: The gateway must identify "Engine Alerts" from the processed (cleaned and normalized) sensor data and route them specifically to the Maintenance Agent for immediate attention.
- Preconditions:
 - Sensor data has been ingested (FR-GW-001), cleaned (FR-GW-003), and normalized (FR-GW-002).
 - Rules for identifying "Engine Alerts" (e.g., specific engine health parameters exceeding thresholds) are defined and configured.
 - The communication channel and recipient details for the Maintenance Agent system are configured.
- Main Flow / Functional Steps:
 - The gateway processes the normalized and cleaned sensor data to detect conditions that constitute an "Engine Alert."
 - Upon detection of an "Engine Alert," the gateway extracts and packages the relevant alert information (e.g., truck ID, alert type, timestamp, specific engine parameter values).
 - The gateway transmits the "Engine Alert" package to the designated Maintenance Agent system or interface.

- The gateway logs the transmission of the alert.

5. FR-GW-005: Route Location Updates to Customer Tracking Agent

- Title: Route Location Updates to Customer Tracking Agent
- Description: The gateway must identify "Location Updates" from the processed (cleaned and normalized) sensor data and route them to the Customer Tracking Agent to provide real-time tracking information.
- Preconditions:
 - Sensor data has been ingested (FR-GW-001), cleaned (FR-GW-003), and normalized (FR-GW-002).
 - Rules for identifying and extracting "Location Updates" are defined.
 - The communication channel and recipient details for the Customer Tracking Agent system are configured.
- Main Flow / Functional Steps:
 - The gateway processes the normalized and cleaned sensor data to extract "Location Updates" (e.g., GPS coordinates, speed, direction).
 - The gateway packages the relevant location information (e.g., truck ID, current location, timestamp).
 - The gateway transmits the "Location Update" package to the designated Customer Tracking Agent system or interface.
 - The gateway logs the transmission of the location update.